

Full-Stack eCommerce Web Architecture

PRODUCTION
ARCHITECTURE

Custom High-Performance Storefront, Django REST API & Microservices

100%
Custom
ZERO SHOPIFY
FEES

< 150ms
API RESPONSE
TIME

Microservices
DOCKER & NGINX

JWT
Auth
SECURE USER
ACCOUNTS

Core System Architecture: Microservices Online Store

Modern Decoupled Architecture (Django REST Framework + Vue 3)

A high-concurrency eCommerce platform built with a decoupled architecture. The frontend provides an app-like reactive shopping experience, while the robust Python Django backend manages inventory transactions, orders, security, and automated background jobs.

Django REST Framework

Vue 3

Vuex Persistent Cart

CeLery & Redis

Docker Compose

Nginx

✓ Product & Category Engine:

Instant category filtering, search autocomplete, and image galleries.

✓ Customer Portal & Auth:

Secure token-based registration, login, profile, and order history tracking.

✓ Reactive Cart Management:

Real-time cart state with persistent local storage and live item counts.

✓ Checkout & Address Validation:

Multi-step shipping address, payment selection, and instant confirmation.

Admin Management & Background Automation

Built-in Business Backoffice (Django Admin & Celery)

Complete control over your online store without third-party recurring subscription fees.

✓ Inventory Control:

Add, edit, stock-track, and organize products and variants with ease.

✓ Order Management:

Inspect paid orders, customer shipping details, and fulfillment tracking.

✓ Asynchronous Processing:

Redis + Celery worker queues for invoice generation and transactional email alerts.

✓ Production Docker Deploy:

Multi-container orchestration with Nginx reverse proxy and SSL.

Why Custom Full-Stack vs. Shopify / WooCommerce

BUSINESS
COMPARISON

Cost, Performance & Scalability Comparison

KEY METRIC	CUSTOM PYTHON/DJANGO + VUE STORE	SHOPIFY / WOOCOMMERCE
Transaction & Monthly Fees	\$0 / month platform fees (Only standard Stripe/PayPal fees)	\$39 - \$399/month + 2% transaction fee penalty on third-party gateways
Speed & Performance	Sub-second Vue 3 SPA loading (instant page transitions)	Often bloated with 20+ plugins, sluggish checkout
Code & Data Ownership	100% full ownership (Full open source repository)	Vendor lock-in (platform can suspend store anytime)
Custom Logic & Features	Unlimited flexibility (Any custom business rules or API integrations)	Requires expensive monthly paid apps for basic features
Security & Stability	Battle-tested Django security (Built-in CSRF, SQLi, XSS protection)	Frequent WordPress plugin zero-day vulnerabilities

Delivery Guarantee & Production Deliverables:

✓ Full Source Code Delivery:	Clean, well-structured frontend and backend repositories.	✓ 1-Click Docker Setup:	Easy `docker compose up` setup for local development or VPS servers.
✓ Payment Gateway Ready:	Clean integration points for Stripe, PayPal, or local processors.	✓ Mobile Responsive UI:	Flawless purchasing experience on iPhone, Android, and tablets.

Ready to launch a high-performance custom store?

Send me a message on Fiverr with your product requirements and let's get your store online!

Delivered with clean architecture & zero monthly platform fees

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